

HLT Diphoton Trigger Efficiency

Production & Verification — 2023 and 2024

CMS low-mass diphoton — tag-and-probe

Compiled July 28, 2026

Scope. This note documents, end to end, how the 2023 and 2024 diphoton HLT leg efficiencies were produced, the fit model used, the category definitions, and the exact nominal efficiency values for every variant, leg, category, and E_T bin. It is a self-contained record for verification. All numbers in Section 8 are read directly from the consolidated result CSVs (`efficiencies_from_txt.csv`) that back the two public result repositories.

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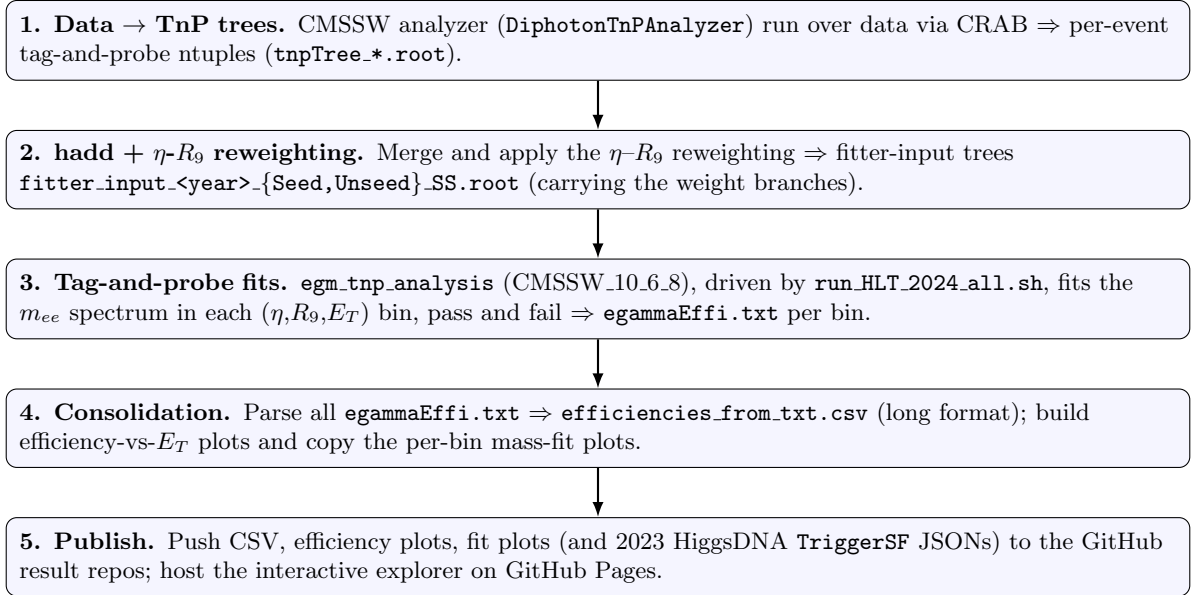
1 Overview

The measurement is a **tag-and-probe** determination of the efficiency of the two legs of the CMS diphoton high-level-trigger (HLT) path, using $Z \rightarrow ee$ electrons as photon probes. Two data-taking years are covered:

- **2023** — two data-taking sub-eras, `preBPix` and `postBPix` (split at the BPix pixel-detector boundary; integrated luminosities 17.96 and 9.68 fb^{-1} respectively).
- **2024** — one era; two independent η - R_9 reweighting choices (`noPresel/withPresel`); the settings carry a placeholder $\mathcal{L} = 109.08 \text{ fb}^{-1}$.

Each year is measured in two trigger-selection variants (*standard* and *fullPath*), two legs (*seeded/unseeded*), five η/R_9 categories, and a set of probe- E_T bins. Results are published at [HLT-diphoton-efficiency-2023-results](#) and [HLT-diphoton-efficiency-2024-results](#), with an interactive cross-year comparison tool at [the GitHub Pages explorer](#).

2 Production workflow



The fit step (3) is the one repeated per bin and per fit model; steps 1–2 are the upstream tree production. The driver script runs, for one **WEIGHT**×**SET**×**VARIANT**, all five categories × both legs, calling the fitter with `--createBins`, `--createHists`, `--doFit` (twice, one per background model), and `--sumUp`.

3 Fit model

For each (η, R_9, E_T) bin the invariant-mass spectrum m_{ee} of tag–probe pairs is fit in the window $[50, 130]$ GeV, separately for probes that *pass* and *fail* the leg requirement. The efficiency is the fitted passing fraction,

$$\varepsilon = \frac{N_{\text{pass}}^{\text{sig}}}{N_{\text{pass}}^{\text{sig}} + N_{\text{fail}}^{\text{sig}}},$$

with N^{sig} the fitted signal yields. The models:

- **Signal:** Breit–Wigner $\otimes$ double-sided Crystal Ball (bwDSCB) — a relativistic Z Breit–Wigner convolved with a two-sided Crystal Ball resolution function.
- **Background:** RooCMSShape (erf×exponential) — the *nominal* model, tag **bwDscbCms**; and **Exponential** — the *alternate*, tag **bwDscbExp**.

The tables in Section 8 quote the nominal (RooCMSShape) result; the exponential-background variant is also present in every CSV and in the explorer, and drives the background systematic. Quoted uncertainties σ_+/σ_- are the asymmetric fit (statistical) errors on ε .

Probe selection (common). tag $E_T > 35$, $|\eta_{\text{tag}}| < 2.5$, tag MVA > -0.6 ; probe MVA > -0.9 , $H/E < 0.08$, $\sigma_{\text{in}\eta}$ and photon-isolation requirements per region, $\text{trkSumPtHollowConeDR03} < 6$. Seeded leg: probe $E_T > 30$; unseeded leg: probe $E_T > 18$ and the tag must have fired the seeded leg. **fullPath** additionally requires **probe_passFullPath**=1.

4 Category & binning definitions

- **Legs:** *seeded* (lead) and *unseeded* (sublead).
- **ECAL region:** EB (barrel, $|\eta| < 1.44$), EE (endcap, $1.57 < |\eta| < 2.5$).
- R_9 **categories** ($R_9 = E_{3\times 3}/E_{\text{RAW}}$): EB {0.50–0.55, 0.55–0.85, >0.85 }; EE {0.90–0.93, >0.93 }.

- **Probe E_T bins:** seeded {35–37, 37–40, 40–45, 45–50, 50–60, 60–70, 70–90, >90}; unseeded starts at 21 with finer low- E_T spacing. The lowest catch-all bin (from 0) is marked [†] in the tables and is excluded from the published efficiency-vs- E_T plots.
- **Variants (2023):** preBPix/postBPix eras $\times$ standard / fullPath.
- **Variants (2024):** η - R_9 reweighting noPresel/withPresel (derived without / with the photon preselection) $\times$ standard / fullPath. The fullPath suffix means the same full-HLT-path requirement in both years.

5 Preselection: exact cuts

Every fit bin applies the same base selection (`cutBase`), plus a per- p_T -bin R_9 category cut (`additionalCuts`) and the leg “pass” flag. The values below are *verbatim* from the 2024 settings files; the EB/EE and seeded/unseeded differences are annotated inline. The `noPresel` and `withPresel` variants use the **identical** `cutBase` — they differ *only* in which η - R_9 weight branch is applied (Section 7).

```
# --- cutBase: base selection applied in EVERY fit bin (2024 settings_*.py) ---
tag_Ele_pt > 35
&& abs(tag_sc_eta) < 2.5
&& tag_mva > -0.6
&& probe_pt > 30          # seeded leg      (unseeded leg: probe_pt > 18)
&& ph_sc_abseta < 1.4442   # EB region      (EE: ph_sc_abseta > 1.566 && < 2.5)
&& probe_mva > -0.9
&& probe_hoe < 0.08
&& ph_full5x5x_r9 > 0.5    # region R9 floor
&& probe_sieie < 0.011     # EB              (EE: probe_sieie < 0.032)
&& probe_phIso_rhoCorr < 3.0
&& probe_trkSumPtHollowConeDR03 < 6.0
&& ( ph_full5x5x_r9 > 0.8 || probe_pfRelIso03_chg_quad*probe_pt < 20
    || probe_pfRelIso03_chg_quad < 0.3 )    # H->gg isolation OR-clause

# unseeded leg additionally requires: && tag_seededPass == 1
# fullPath variant additionally adds: && probe_passFullPath == 1

# per-pt-bin R9 category cut (additionalCuts), e.g. EB 0.50-0.55 category:
#   ph_full5x5x_r9 > 0.50 && ph_full5x5x_r9 < 0.55
# pass flag (defines the numerator): probe_passSeed == 1   (unseeded: probe_passUnseed == 1)
```

6 TnP framework: how the fits are run

The fits use the CMS EGamma POG `egm.tnp.analysis` toolkit (CMSSW_10_6_8). For each category $\times$ leg the driver calls `tnpEGM.fitter.py` through a fixed sequence of stages:

```
# run_HLT_2024_all.sh <WEIGHT> <SET> <VARIANT>   WEIGHT=noPresel|withPresel  SET=normal|fullPath
FITTER="python tnpEGM.fitter.py"
for CAT in EB_R9_0p50to0p55 EB_R9_0p55to0p85 EB_R9gt0p85 EE_R9_0p90to0p93 EE_R9gt0p93; do
  for LEG in seed unseed; do
    CFG=settings_${CAT}_${LEG}_HLT_2024_${WEIGHT}_${SUFFIX}.py
    FLAG=passingHLT_${LEG}_${CAT}_2024
    $FITTER $CFG --flag $FLAG --checkBins          # sanity-check the bin definitions
    $FITTER $CFG --flag $FLAG --createBins          # build (ph_sc_abseta, ph_sc_et) bins
    $FITTER $CFG --flag $FLAG --createHists         # fill pass/fail pair_mass hists (WEIGHTED)
    $FITTER $CFG --flag $FLAG --doFit --bwDscb      # BWxDSCB signal + RooCMSShape bkg (nominal)
    $FITTER $CFG --flag $FLAG --doFit --bwDscbExp   # BWxDSCB signal + Exponential bkg (alternate)
    $FITTER $CFG --flag $FLAG --sumUp              # write egammaEffi.txt (eff, +err, -err / bin)
  done
done
```

`--createHists` fills, for each (η, E_T) bin, a *passing* and a *failing* m_{ee} histogram (`pair_mass`, 80 bins, [50, 130] GeV) using `TTree::Draw` with the per-event weight (Section 7); “passing” is defined by the flag `probe_passSeed==1` (or `probe_passUnseed==1`). `--doFit` fits the signal+background model to both; `--sumUp` writes `egammaEffi.txt` with columns `bwDscbCms_{eff,elo,ehi}` and `bwDscbExp_{eff,elo,ehi}`. Signal-model parameter initialisation:

```

# tnpParBWdSCBFit (signal = Breit-Wigner (x) double-sided Crystal Ball; P=passing, F=failing)
meanP[0.0,-3.0,3.0]   sigmaP[1.5,0.3,5.0]   meanF[0.0,-3.0,3.0]   sigmaF[1.5,0.3,5.0]
alpha1P[1.5,0.5,5.0]  n1P[2.0,1.0,10.0]      alpha2P[1.5,0.5,5.0]  n2P[2.0,1.0,10.0]
alpha1F[1.5,0.5,5.0]  n1F[2.0,1.0,10.0]      alpha2F[1.5,0.5,5.0]  n2F[2.0,1.0,10.0]
# nominal background = RooCMSShape (erf x exp):
acmsP[60.,50.,100.]  betaP[0.05,0.01,0.08]  gammaP[0.1,0,1]  peakP[90.0]      # + acmsF/betaF/gammaF/peakF
# alternate background (tnpParBWdSCBExpFit) replaces the CMSShape params with an exponential:
alphaP[0.,-2.,1.]   alphaF[0.,-2.,1.]
# fit variable: pair_mass, 80 bins in [50,130] GeV (tnpEGM_fitter.py)
# efficiency = N_sig_pass / (N_sig_pass + N_sig_fail)

```

7 η - R_9 reweighting — how it enters the workflow

The η - R_9 reweighting is **not** a selection: it is a per-event multiplicative weight, looked up from an R_9 -vs- $|\eta|$ ratio map (the “MC R_9 -vs- $|\eta|$ correction” stored under `MC_R9eta_weights/`), written as a branch in the fitter-input tree, and applied by the fitter when it fills the mass histograms that are fit. The four concrete steps — map, per-event lookup, settings selection, and fitter application — are:

```

# (1) Two MC-derived R9-vs-|eta| correction maps (TH2, key "R9_vs_eta"):
# no_presel : MC_R9eta_weights/2024/2024_R9_vs_eta_ratio_vf.root (event <w> ~ 1.7)
# with_presel : MC_R9eta_weights/2024/2024_R9_vs_eta_ratio_2024_presel_vf.root (event <w> ~ 1)
# no_presel ratio is built over ALL probes (junk-tail driven);
# with_presel ratio is built with the 2023-style preselection applied (clean).

R9_EDGES = [0, 0.5, 0.55, 0.6, 0.65, 0.7, 0.74, 0.76, 0.78, 0.8, 0.82, 0.84, 0.86, 0.88,
            0.9, 0.91, 0.92, 0.93, 0.94, 0.95, 0.96, 0.97, 0.98, 0.99, 1.05, 2.0] # 25 R9 bins
ETA_EDGES = [0, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4442, 1.566, 1.7, 1.8, 2.0,
            2.2, 2.3, 2.5] # 16 |eta| bins

# (2) per-event lookup in make_fitter_input_2024.ipynb (each probe -> one weight):
def lookup_r9eta_weight(weights_2d, r9_arr, eta_arr):
    ix = clip(digitize(r9_arr, R9_EDGES) - 1, 0, nR9 - 1)
    iy = clip(digitize(abs(eta_arr), ETA_EDGES) - 1, 0, nEta - 1)
    return weights_2d[ix, iy] # -> r9etaWeight_{no,with}_presel -> weight_{no,with}_presel
# these are written as branches into fitter_input_2024_{Seed,Unseed}_SS.root

# (3) the fitter settings file SELECTS which variant to apply:
tnpSample('data_2024_noPresel_seed_EB_R9_0p50to0p55', _input,
         lumi=109.08, weight='weight_no_presel') # withPresel -> weight='weight_with_presel'

# (4) the fitter APPLIES it when filling the pass/fail mass histograms (libPython/rootUtils.py):
if not sample.weight is None:
    cutPass = '( %s && %s ) * %s ' % (cuts, flag, sample.weight) # weighted PASS histogram
    cutFail = '( %s && %s ) * %s ' % (cuts, notflag, sample.weight) # weighted FAIL histogram

```

Why there are two variants. `no_presel` builds the R_9 -vs- $|\eta|$ ratio over all probes (its event weight averages ~ 1.7 , driven by the low- R_9 junk tail); `with_presel` builds the ratio after applying the 2023-style preselection (weights average ~ 1 , cleaner). Because both weight branches live in the *same* input tree and the cutBase is identical, the only thing that changes between the `etaR9rw_noPresel` and `etaR9rw_withPresel` results is the `weight=` argument in the settings file — which is exactly what makes them a controlled comparison of the reweighting choice. (The input trees are the `_SS` files, i.e. the photon scale/smearing “SS” correction is already applied upstream; the weight here is the η - R_9 factor only.)

8 Exact efficiency values

Nominal efficiencies $\varepsilon_{-\sigma_-}^{+\sigma_+}$ (`bwDSCB` $\otimes$ `RooCMSShape`), read directly from the result CSVs. Cross-check totals: 2024 — 420 nominal points, $\varepsilon \in [0.7455, 0.9999]$; 2023 — 420 nominal points, $\varepsilon \in [0.6451, 1.0000]$. Rows marked † are the lowest catch-all E_T bin (excluded from the published curves).

8.1 2024

Table 1: 2024 – η - R_9 rwt, no presel – standard (SS) – seeded (lead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–35 [†]	0.8288 ^{+0.0060} _{–0.0061}	0.8730 ^{+0.0011} _{–0.0011}	0.8724 ^{+0.0010} _{–0.0010}	0.8394 ^{+0.0045} _{–0.0046}	0.8924 ^{+0.0010} _{–0.0010}
35–37	0.9022 ^{+0.0032} _{–0.0020}	0.9360 ^{+0.0010} _{–0.0010}	0.9175 ^{+0.0010} _{–0.0010}	0.9064 ^{+0.0046} _{–0.0013}	0.9439 ^{+0.0010} _{–0.0010}
37–40	0.9103 ^{+0.0013} _{–0.0013}	0.9414 ^{+0.0010} _{–0.0010}	0.9273 ^{+0.0010} _{–0.0010}	0.9196 ^{+0.0010} _{–0.0010}	0.9478 ^{+0.0010} _{–0.0010}
40–45	0.9220 ^{+0.0016} _{–0.0016}	0.9493 ^{+0.0010} _{–0.0010}	0.9374 ^{+0.0010} _{–0.0010}	0.9361 ^{+0.0010} _{–0.0010}	0.9592 ^{+0.0010} _{–0.0010}
45–50	0.9282 ^{+0.0016} _{–0.0016}	0.9515 ^{+0.0010} _{–0.0010}	0.9450 ^{+0.0010} _{–0.0010}	0.9464 ^{+0.0010} _{–0.0010}	0.9647 ^{+0.0010} _{–0.0010}
50–60	0.9215 ^{+0.0076} _{–0.0071}	0.9478 ^{+0.0010} _{–0.0010}	0.9507 ^{+0.0010} _{–0.0010}	0.9429 ^{+0.0028} _{–0.0050}	0.9667 ^{+0.0010} _{–0.0017}
60–70	0.9344 ^{+0.0157} _{–0.0196}	0.9476 ^{+0.0010} _{–0.0010}	0.9544 ^{+0.0010} _{–0.0010}	0.9528 ^{+0.0056} _{–0.0042}	0.9752 ^{+0.0036} _{–0.0041}
70–90	0.9379 ^{+0.0127} _{–0.0145}	0.9664 ^{+0.0106} _{–0.0042}	0.9741 ^{+0.0014} _{–0.0017}	0.9708 ^{+0.0058} _{–0.0042}	0.9798 ^{+0.0010} _{–0.0010}
90– ∞	0.9554 ^{+0.0216} _{–0.0218}	0.9760 ^{+0.0028} _{–0.0026}	0.9803 ^{+0.0010} _{–0.0010}	0.9757 ^{+0.0030} _{–0.0026}	0.9827 ^{+0.0010} _{–0.0010}

Table 2: 2024 – η - R_9 rwt, no presel – standard (SS) – unseeded (sublead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–21 [†]	0.7463 ^{+0.0096} _{–0.0082}	0.8608 ^{+0.0023} _{–0.0023}	0.8924 ^{+0.0026} _{–0.0026}	0.8048 ^{+0.0080} _{–0.0085}	0.7455 ^{+0.0010} _{–0.0010}
21–25	0.8552 ^{+0.0039} _{–0.0035}	0.9096 ^{+0.0028} _{–0.0043}	0.9198 ^{+0.0010} _{–0.0010}	0.8587 ^{+0.0048} _{–0.0089}	0.9101 ^{+0.0019} _{–0.0019}
25–28	0.8964 ^{+0.0139} _{–0.0177}	0.9473 ^{+0.0020} _{–0.0021}	0.9439 ^{+0.0010} _{–0.0012}	0.8994 ^{+0.0174} _{–0.0114}	0.9174 ^{+0.0010} _{–0.0010}
28–31	0.9219 ^{+0.0091} _{–0.0093}	0.9570 ^{+0.0017} _{–0.0017}	0.9599 ^{+0.0010} _{–0.0010}	0.9048 ^{+0.0085} _{–0.0088}	0.9474 ^{+0.0010} _{–0.0010}
31–35	0.9427 ^{+0.0039} _{–0.0040}	0.9676 ^{+0.0010} _{–0.0010}	0.9651 ^{+0.0010} _{–0.0010}	0.9217 ^{+0.0031} _{–0.0027}	0.9571 ^{+0.0010} _{–0.0010}
35–40	0.9375 ^{+0.0013} _{–0.0010}	0.9657 ^{+0.0010} _{–0.0010}	0.9688 ^{+0.0010} _{–0.0010}	0.9345 ^{+0.0010} _{–0.0010}	0.9642 ^{+0.0010} _{–0.0010}
40–45	0.9491 ^{+0.0010} _{–0.0010}	0.9717 ^{+0.0010} _{–0.0010}	0.9748 ^{+0.0010} _{–0.0010}	0.9483 ^{+0.0010} _{–0.0010}	0.9749 ^{+0.0010} _{–0.0010}
45–50	0.9528 ^{+0.0015} _{–0.0016}	0.9716 ^{+0.0010} _{–0.0010}	0.9759 ^{+0.0010} _{–0.0010}	0.9519 ^{+0.0010} _{–0.0010}	0.9791 ^{+0.0010} _{–0.0010}
50–60	0.9434 ^{+0.0057} _{–0.0026}	0.9648 ^{+0.0010} _{–0.0010}	0.9746 ^{+0.0010} _{–0.0010}	0.9488 ^{+0.0027} _{–0.0054}	0.9820 ^{+0.0010} _{–0.0016}
60–70	0.9517 ^{+0.0144} _{–0.0264}	0.9636 ^{+0.0014} _{–0.0014}	0.9728 ^{+0.0010} _{–0.0010}	0.9566 ^{+0.0024} _{–0.0025}	0.9862 ^{+0.0024} _{–0.0024}
70–90	0.9513 ^{+0.0114} _{–0.0114}	0.9760 ^{+0.0056} _{–0.0016}	0.9829 ^{+0.0012} _{–0.0012}	0.9755 ^{+0.0039} _{–0.0090}	0.9841 ^{+0.0010} _{–0.0010}
90– ∞	0.9484 ^{+0.0226} _{–0.0229}	0.9807 ^{+0.0037} _{–0.0037}	0.9837 ^{+0.0010} _{–0.0010}	0.9744 ^{+0.0020} _{–0.0022}	0.9901 ^{+0.0010} _{–0.0010}

Table 3: 2024 – η - R_9 rwt, with presel – standard (SS) – seeded (lead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–35 [†]	0.8162 ^{+0.0045} _{–0.0046}	0.8669 ^{+0.0010} _{–0.0010}	0.8732 ^{+0.0010} _{–0.0010}	0.8367 ^{+0.0052} _{–0.0054}	0.8985 ^{+0.0010} _{–0.0010}
35–37	0.8960 ^{+0.0025} _{–0.0025}	0.9348 ^{+0.0033} _{–0.0033}	0.9178 ^{+0.0010} _{–0.0010}	0.9133 ^{+0.0032} _{–0.0030}	0.9438 ^{+0.0010} _{–0.0010}
37–40	0.9050 ^{+0.0010} _{–0.0011}	0.9395 ^{+0.0010} _{–0.0010}	0.9273 ^{+0.0010} _{–0.0010}	0.9177 ^{+0.0010} _{–0.0010}	0.9476 ^{+0.0010} _{–0.0010}
40–45	0.9157 ^{+0.0010} _{–0.0031}	0.9470 ^{+0.0010} _{–0.0010}	0.9375 ^{+0.0010} _{–0.0010}	0.9343 ^{+0.0010} _{–0.0010}	0.9587 ^{+0.0010} _{–0.0010}
45–50	0.9214 ^{+0.0013} _{–0.0013}	0.9495 ^{+0.0010} _{–0.0010}	0.9449 ^{+0.0010} _{–0.0010}	0.9448 ^{+0.0010} _{–0.0010}	0.9642 ^{+0.0010} _{–0.0010}
50–60	0.9163 ^{+0.0053} _{–0.0031}	0.9466 ^{+0.0010} _{–0.0011}	0.9486 ^{+0.0010} _{–0.0010}	0.9481 ^{+0.0030} _{–0.0056}	0.9664 ^{+0.0010} _{–0.0018}
60–70	0.9323 ^{+0.0126} _{–0.0156}	0.9523 ^{+0.0010} _{–0.0010}	0.9537 ^{+0.0015} _{–0.0015}	0.9537 ^{+0.0051} _{–0.0068}	0.9764 ^{+0.0038} _{–0.0048}
70–90	0.9345 ^{+0.0088} _{–0.0088}	0.9647 ^{+0.0082} _{–0.0055}	0.9730 ^{+0.0036} _{–0.0012}	0.9709 ^{+0.0063} _{–0.0050}	0.9793 ^{+0.0014} _{–0.0013}
90– ∞	0.9511 ^{+0.0162} _{–0.0162}	0.9772 ^{+0.0018} _{–0.0089}	0.9804 ^{+0.0010} _{–0.0010}	0.9753 ^{+0.0034} _{–0.0024}	0.9825 ^{+0.0010} _{–0.0010}

Table 4: 2024 – η - R_9 rwt, with presel – standard (SS) – unseeded (sublead) leg. Nominal $\varepsilon_{-\sigma_-}^{+\sigma_+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–21 [†]	0.7488 ^{+0.0165} _{–0.0081}	0.8519 ^{+0.0018} _{–0.0019}	0.8114 ^{+0.0010} _{–0.0010}	0.8025 ^{+0.0101} _{–0.0101}	0.8365 ^{+0.0042} _{–0.0042}
21–25	0.8515 ^{+0.0203} _{–0.0025}	0.9101 ^{+0.0012} _{–0.0032}	0.9044 ^{+0.0012} _{–0.0010}	0.8443 ^{+0.0272} _{–0.0074}	0.8806 ^{+0.0026} _{–0.0026}
25–28	0.8992 ^{+0.0075} _{–0.0139}	0.9464 ^{+0.0018} _{–0.0019}	0.9511 ^{+0.0014} _{–0.0014}	0.8516 ^{+0.0062} _{–0.0042}	0.9043 ^{+0.0011} _{–0.0011}
28–31	0.9208 ^{+0.0010} _{–0.0011}	0.9563 ^{+0.0013} _{–0.0013}	0.9602 ^{+0.0010} _{–0.0010}	0.9078 ^{+0.0086} _{–0.0091}	0.9482 ^{+0.0011} _{–0.0011}
31–35	0.9380 ^{+0.0034} _{–0.0034}	0.9671 ^{+0.0025} _{–0.0010}	0.9650 ^{+0.0010} _{–0.0010}	0.9209 ^{+0.0040} _{–0.0040}	0.9529 ^{+0.0010} _{–0.0010}
35–40	0.9386 ^{+0.0010} _{–0.0010}	0.9653 ^{+0.0010} _{–0.0010}	0.9677 ^{+0.0010} _{–0.0010}	0.9339 ^{+0.0010} _{–0.0010}	0.9640 ^{+0.0010} _{–0.0010}
40–45	0.9495 ^{+0.0010} _{–0.0010}	0.9714 ^{+0.0010} _{–0.0010}	0.9748 ^{+0.0010} _{–0.0010}	0.9480 ^{+0.0010} _{–0.0012}	0.9749 ^{+0.0010} _{–0.0010}
45–50	0.9531 ^{+0.0011} _{–0.0010}	0.9714 ^{+0.0010} _{–0.0010}	0.9758 ^{+0.0010} _{–0.0010}	0.9521 ^{+0.0010} _{–0.0010}	0.9790 ^{+0.0010} _{–0.0010}
50–60	0.9446 ^{+0.0033} _{–0.0021}	0.9641 ^{+0.0010} _{–0.0010}	0.9742 ^{+0.0010} _{–0.0010}	0.9570 ^{+0.0031} _{–0.0066}	0.9819 ^{+0.0010} _{–0.0010}
60–70	0.9515 ^{+0.0184} _{–0.0188}	0.9634 ^{+0.0010} _{–0.0010}	0.9723 ^{+0.0010} _{–0.0010}	0.9526 ^{+0.0034} _{–0.0030}	0.9881 ^{+0.0031} _{–0.0031}
70–90	0.9527 ^{+0.0064} _{–0.0073}	0.9769 ^{+0.0051} _{–0.0044}	0.9825 ^{+0.0014} _{–0.0014}	0.9739 ^{+0.0072} _{–0.0103}	0.9839 ^{+0.0013} _{–0.0010}
90– ∞	0.9448 ^{+0.0185} _{–0.0184}	0.9774 ^{+0.0025} _{–0.0014}	0.9835 ^{+0.0010} _{–0.0010}	0.9740 ^{+0.0039} _{–0.0039}	0.9903 ^{+0.0010} _{–0.0010}

Table 5: 2024 – η - R_9 rwt, no presel – fullPath – seeded (lead) leg. Nominal $\varepsilon_{-\sigma_-}^{+\sigma_+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–35 [†]	0.8790 ^{+0.0047} _{–0.0054}	0.8885 ^{+0.0010} _{–0.0010}	0.8978 ^{+0.0010} _{–0.0010}	0.9121 ^{+0.0092} _{–0.0043}	0.9299 ^{+0.0022} _{–0.0010}
35–37	0.9494 ^{+0.0014} _{–0.0014}	0.9579 ^{+0.0010} _{–0.0010}	0.9412 ^{+0.0010} _{–0.0010}	0.9658 ^{+0.0011} _{–0.0011}	0.9713 ^{+0.0010} _{–0.0010}
37–40	0.9582 ^{+0.0010} _{–0.0010}	0.9638 ^{+0.0010} _{–0.0010}	0.9476 ^{+0.0010} _{–0.0010}	0.9715 ^{+0.0010} _{–0.0010}	0.9733 ^{+0.0010} _{–0.0010}
40–45	0.9615 ^{+0.0010} _{–0.0010}	0.9673 ^{+0.0010} _{–0.0010}	0.9534 ^{+0.0010} _{–0.0010}	0.9784 ^{+0.0010} _{–0.0010}	0.9765 ^{+0.0010} _{–0.0010}
45–50	0.9640 ^{+0.0013} _{–0.0012}	0.9694 ^{+0.0010} _{–0.0010}	0.9601 ^{+0.0010} _{–0.0010}	0.9854 ^{+0.0010} _{–0.0010}	0.9778 ^{+0.0010} _{–0.0010}
50–60	0.9665 ^{+0.0043} _{–0.0047}	0.9719 ^{+0.0010} _{–0.0010}	0.9673 ^{+0.0010} _{–0.0010}	0.9849 ^{+0.0010} _{–0.0010}	0.9778 ^{+0.0010} _{–0.0010}
60–70	0.9770 ^{+0.0064} _{–0.0049}	0.9803 ^{+0.0010} _{–0.0010}	0.9762 ^{+0.0010} _{–0.0010}	0.9875 ^{+0.0021} _{–0.0020}	0.9856 ^{+0.0010} _{–0.0010}
70–90	0.9793 ^{+0.0070} _{–0.0071}	0.9853 ^{+0.0014} _{–0.0016}	0.9865 ^{+0.0010} _{–0.0010}	0.9931 ^{+0.0019} _{–0.0019}	0.9871 ^{+0.0010} _{–0.0010}
90– ∞	0.9870 ^{+0.0073} _{–0.0110}	0.9932 ^{+0.0010} _{–0.0014}	0.9933 ^{+0.0010} _{–0.0010}	0.9979 ^{+0.0010} _{–0.0010}	0.9900 ^{+0.0010} _{–0.0010}

Table 6: 2024 – η - R_9 rwt, no presel – fullPath – unseeded (sublead) leg. Nominal $\varepsilon_{-\sigma_-}^{+\sigma_+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–21 [†]	0.9897 ^{+0.0070} _{–0.0072}	0.9925 ^{+0.0010} _{–0.0010}	0.9927 ^{+0.0010} _{–0.0010}	0.9827 ^{+0.0090} _{–0.0023}	0.9815 ^{+0.0010} _{–0.0010}
21–25	0.9968 ^{+0.0029} _{–0.0043}	0.9982 ^{+0.0010} _{–0.0010}	0.9966 ^{+0.0010} _{–0.0010}	0.9956 ^{+0.0053} _{–0.0012}	0.9938 ^{+0.0010} _{–0.0010}
25–28	0.9985 ^{+0.0010} _{–0.0010}	0.9989 ^{+0.0010} _{–0.0010}	0.9985 ^{+0.0010} _{–0.0010}	0.9971 ^{+0.0017} _{–0.0010}	0.9979 ^{+0.0010} _{–0.0010}
28–31	0.9985 ^{+0.0012} _{–0.0010}	0.9992 ^{+0.0010} _{–0.0010}	0.9993 ^{+0.0010} _{–0.0010}	0.9979 ^{+0.0010} _{–0.0010}	0.9988 ^{+0.0010} _{–0.0010}
31–35	0.9994 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9996 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
35–40	0.9994 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}
40–45	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
45–50	0.9997 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
50–60	0.9993 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9992 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
60–70	0.9982 ^{+0.0011} _{–0.0011}	0.9992 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9993 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
70–90	0.9976 ^{+0.0028} _{–0.0028}	0.9993 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9992 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
90– ∞	0.9930 ^{+0.0032} _{–0.0032}	0.9993 ^{+0.0010} _{–0.0010}	0.9996 ^{+0.0010} _{–0.0010}	0.9994 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}

Table 7: 2024 – η - R_9 rwt, with presel – fullPath – seeded (lead) leg. Nominal $\varepsilon_{-\sigma_-}^{+\sigma_+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–35 [†]	0.8685 ^{+0.0038} _{–0.0040}	0.8927 ^{+0.0010} _{–0.0010}	0.8965 ^{+0.0010} _{–0.0010}	0.9113 ^{+0.0045} _{–0.0132}	0.9287 ^{+0.0010} _{–0.0010}
35–37	0.9425 ^{+0.0011} _{–0.0012}	0.9563 ^{+0.0010} _{–0.0010}	0.9412 ^{+0.0010} _{–0.0010}	0.9643 ^{+0.0016} _{–0.0011}	0.9700 ^{+0.0010} _{–0.0010}
37–40	0.9519 ^{+0.0010} _{–0.0010}	0.9623 ^{+0.0010} _{–0.0010}	0.9476 ^{+0.0010} _{–0.0010}	0.9704 ^{+0.0010} _{–0.0010}	0.9728 ^{+0.0010} _{–0.0010}
40–45	0.9548 ^{+0.0010} _{–0.0010}	0.9655 ^{+0.0010} _{–0.0010}	0.9535 ^{+0.0010} _{–0.0010}	0.9770 ^{+0.0010} _{–0.0010}	0.9761 ^{+0.0010} _{–0.0010}
45–50	0.9567 ^{+0.0012} _{–0.0010}	0.9677 ^{+0.0010} _{–0.0010}	0.9602 ^{+0.0010} _{–0.0010}	0.9838 ^{+0.0010} _{–0.0010}	0.9771 ^{+0.0010} _{–0.0010}
50–60	0.9593 ^{+0.0046} _{–0.0037}	0.9688 ^{+0.0010} _{–0.0010}	0.9685 ^{+0.0010} _{–0.0010}	0.9828 ^{+0.0052} _{–0.0019}	0.9771 ^{+0.0010} _{–0.0010}
60–70	0.9706 ^{+0.0038} _{–0.0060}	0.9782 ^{+0.0026} _{–0.0026}	0.9762 ^{+0.0010} _{–0.0010}	0.9876 ^{+0.0029} _{–0.0025}	0.9848 ^{+0.0026} _{–0.0019}
70–90	0.9753 ^{+0.0050} _{–0.0049}	0.9842 ^{+0.0024} _{–0.0024}	0.9864 ^{+0.0010} _{–0.0010}	0.9928 ^{+0.0019} _{–0.0019}	0.9862 ^{+0.0011} _{–0.0010}
90– ∞	0.9830 ^{+0.0092} _{–0.0092}	0.9928 ^{+0.0010} _{–0.0013}	0.9933 ^{+0.0010} _{–0.0010}	0.9978 ^{+0.0013} _{–0.0013}	0.9896 ^{+0.0010} _{–0.0010}

Table 8: 2024 – η - R_9 rwt, with presel – fullPath – unseeded (sublead) leg. Nominal $\varepsilon_{-\sigma_-}^{+\sigma_+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–21 [†]	0.9886 ^{+0.0048} _{–0.0058}	0.9924 ^{+0.0024} _{–0.0012}	0.9917 ^{+0.0010} _{–0.0010}	0.9790 ^{+0.0099} _{–0.0028}	0.9800 ^{+0.0010} _{–0.0010}
21–25	0.9963 ^{+0.0017} _{–0.0017}	0.9975 ^{+0.0011} _{–0.0011}	0.9972 ^{+0.0010} _{–0.0010}	0.9910 ^{+0.0010} _{–0.0013}	0.9926 ^{+0.0014} _{–0.0010}
25–28	0.9980 ^{+0.0010} _{–0.0010}	0.9988 ^{+0.0010} _{–0.0010}	0.9984 ^{+0.0010} _{–0.0010}	0.9968 ^{+0.0010} _{–0.0010}	0.9974 ^{+0.0010} _{–0.0010}
28–31	0.9992 ^{+0.0010} _{–0.0014}	0.9992 ^{+0.0010} _{–0.0010}	0.9991 ^{+0.0010} _{–0.0010}	0.9979 ^{+0.0010} _{–0.0010}	0.9988 ^{+0.0010} _{–0.0010}
31–35	0.9998 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9996 ^{+0.0010} _{–0.0010}	0.9989 ^{+0.0010} _{–0.0010}	0.9994 ^{+0.0010} _{–0.0010}
35–40	0.9994 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}
40–45	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
45–50	0.9997 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
50–60	0.9995 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9993 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
60–70	0.9981 ^{+0.0010} _{–0.0010}	0.9992 ^{+0.0010} _{–0.0010}	0.9994 ^{+0.0010} _{–0.0010}	0.9993 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
70–90	0.9978 ^{+0.0015} _{–0.0015}	0.9994 ^{+0.0010} _{–0.0010}	0.9996 ^{+0.0010} _{–0.0010}	0.9992 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
90– ∞	0.9906 ^{+0.0040} _{–0.0040}	0.9991 ^{+0.0010} _{–0.0010}	0.9996 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}

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Table 9: 2023 – preBPix – standard (SS) – seeded (lead) leg. Nominal $\varepsilon_{-\sigma_-}^{+\sigma_+}$ (bwDSCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–35 [†]	0.8269 ^{+0.0068} _{–0.0068}	0.8786 ^{+0.0037} _{–0.0037}	0.8840 ^{+0.0014} _{–0.0014}	0.7923 ^{+0.0202} _{–0.0244}	0.8353 ^{+0.0046} _{–0.0047}
35–37	0.9156 ^{+0.0079} _{–0.0079}	0.9417 ^{+0.0011} _{–0.0012}	0.9151 ^{+0.0010} _{–0.0010}	0.9472 ^{+0.0137} _{–0.0136}	0.9510 ^{+0.0032} _{–0.0028}
37–40	0.9125 ^{+0.0037} _{–0.0038}	0.9509 ^{+0.0045} _{–0.0045}	0.9308 ^{+0.0010} _{–0.0010}	0.9404 ^{+0.0034} _{–0.0036}	0.9600 ^{+0.0010} _{–0.0010}
40–45	0.9292 ^{+0.0026} _{–0.0026}	0.9577 ^{+0.0010} _{–0.0010}	0.9416 ^{+0.0010} _{–0.0010}	0.9565 ^{+0.0025} _{–0.0024}	0.9673 ^{+0.0010} _{–0.0010}
45–50	0.9375 ^{+0.0034} _{–0.0034}	0.9578 ^{+0.0010} _{–0.0010}	0.9473 ^{+0.0010} _{–0.0010}	0.9626 ^{+0.0043} _{–0.0030}	0.9733 ^{+0.0010} _{–0.0013}
50–60	0.9308 ^{+0.0064} _{–0.0059}	0.9514 ^{+0.0038} _{–0.0018}	0.9485 ^{+0.0010} _{–0.0010}	0.9661 ^{+0.0053} _{–0.0066}	0.9728 ^{+0.0044} _{–0.0013}
60–70	0.9834 ^{+0.0128} _{–0.0129}	0.9591 ^{+0.0104} _{–0.0128}	0.9544 ^{+0.0014} _{–0.0014}	0.9826 ^{+0.0051} _{–0.0069}	0.9824 ^{+0.0041} _{–0.0053}
70–90	0.9845 ^{+0.0117} _{–0.0117}	0.9786 ^{+0.0061} _{–0.0062}	0.9752 ^{+0.0097} _{–0.0045}	0.9874 ^{+0.0069} _{–0.0070}	0.9869 ^{+0.0052} _{–0.0052}
90– ∞	0.8836 ^{+0.0570} _{–0.0824}	0.9764 ^{+0.0096} _{–0.0044}	0.9847 ^{+0.0010} _{–0.0010}	0.9912 ^{+0.0046} _{–0.0046}	0.9886 ^{+0.0023} _{–0.0024}

Table 10: 2023 – preBPix – standard (SS) – unseeded (sublead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–21 †	0.7613 $^{+0.0367}_{-0.0330}$	0.8132 $^{+0.0103}_{-0.0102}$	0.8759 $^{+0.0032}_{-0.0032}$	0.7503 $^{+0.0865}_{-0.1701}$	0.7679 $^{+0.0130}_{-0.0131}$
21–25	0.9319 $^{+0.0167}_{-0.0412}$	0.9337 $^{+0.0255}_{-0.0025}$	0.9410 $^{+0.0026}_{-0.0014}$	0.8804 $^{+0.0250}_{-0.0576}$	0.9115 $^{+0.0017}_{-0.0075}$
25–28	0.9246 $^{+0.0063}_{-0.0093}$	0.9585 $^{+0.0042}_{-0.0041}$	0.9639 $^{+0.0022}_{-0.0022}$	0.9455 $^{+0.0173}_{-0.0204}$	0.9457 $^{+0.0059}_{-0.0044}$
28–31	0.9400 $^{+0.0177}_{-0.0175}$	0.9787 $^{+0.0050}_{-0.0046}$	0.9800 $^{+0.0017}_{-0.0017}$	0.9634 $^{+0.0122}_{-0.0219}$	0.9704 $^{+0.0032}_{-0.0094}$
31–35	0.9535 $^{+0.0029}_{-0.0030}$	0.9814 $^{+0.0022}_{-0.0024}$	0.9809 $^{+0.0010}_{-0.0010}$	0.9623 $^{+0.0078}_{-0.0080}$	0.9736 $^{+0.0027}_{-0.0024}$
35–40	0.9556 $^{+0.0052}_{-0.0065}$	0.9777 $^{+0.0013}_{-0.0010}$	0.9814 $^{+0.0010}_{-0.0010}$	0.9633 $^{+0.0011}_{-0.0024}$	0.9830 $^{+0.0010}_{-0.0010}$
40–45	0.9604 $^{+0.0034}_{-0.0022}$	0.9816 $^{+0.0010}_{-0.0010}$	0.9846 $^{+0.0010}_{-0.0010}$	0.9754 $^{+0.0018}_{-0.0019}$	0.9867 $^{+0.0010}_{-0.0010}$
45–50	0.9693 $^{+0.0027}_{-0.0027}$	0.9813 $^{+0.0010}_{-0.0010}$	0.9846 $^{+0.0010}_{-0.0010}$	0.9788 $^{+0.0057}_{-0.0034}$	0.9885 $^{+0.0010}_{-0.0010}$
50–60	0.9568 $^{+0.0049}_{-0.0051}$	0.9732 $^{+0.0010}_{-0.0010}$	0.9805 $^{+0.0010}_{-0.0010}$	0.9779 $^{+0.0052}_{-0.0050}$	0.9905 $^{+0.0010}_{-0.0010}$
60–70	0.9743 $^{+0.0204}_{-0.0203}$	0.9821 $^{+0.0068}_{-0.0090}$	0.9800 $^{+0.0074}_{-0.0041}$	0.9828 $^{+0.0074}_{-0.0079}$	0.9915 $^{+0.0019}_{-0.0019}$
70–90	0.9923 $^{+0.0046}_{-0.0046}$	0.9918 $^{+0.0033}_{-0.0033}$	0.9833 $^{+0.0010}_{-0.0010}$	0.9927 $^{+0.0055}_{-0.0055}$	0.9917 $^{+0.0017}_{-0.0017}$
90– ∞	0.9434 $^{+0.0325}_{-0.0660}$	0.9825 $^{+0.0085}_{-0.0042}$	0.9885 $^{+0.0010}_{-0.0010}$	0.9884 $^{+0.0052}_{-0.0052}$	0.9948 $^{+0.0015}_{-0.0016}$

Table 11: 2023 – postBPix – standard (SS) – seeded (lead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwDSCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–35 †	0.7581 $^{+0.0096}_{-0.0101}$	0.8006 $^{+0.0030}_{-0.0031}$	0.8623 $^{+0.0012}_{-0.0010}$	0.8314 $^{+0.0050}_{-0.0037}$	0.8669 $^{+0.0022}_{-0.0025}$
35–37	0.8818 $^{+0.0012}_{-0.0012}$	0.9237 $^{+0.0010}_{-0.0010}$	0.9081 $^{+0.0010}_{-0.0010}$	0.9090 $^{+0.0069}_{-0.0060}$	0.9393 $^{+0.0018}_{-0.0010}$
37–40	0.9013 $^{+0.0021}_{-0.0021}$	0.9357 $^{+0.0010}_{-0.0010}$	0.9198 $^{+0.0010}_{-0.0010}$	0.9261 $^{+0.0019}_{-0.0020}$	0.9543 $^{+0.0010}_{-0.0010}$
40–45	0.9176 $^{+0.0010}_{-0.0010}$	0.9468 $^{+0.0010}_{-0.0010}$	0.9332 $^{+0.0010}_{-0.0010}$	0.9403 $^{+0.0013}_{-0.0014}$	0.9631 $^{+0.0010}_{-0.0010}$
45–50	0.9274 $^{+0.0027}_{-0.0056}$	0.9523 $^{+0.0010}_{-0.0010}$	0.9420 $^{+0.0010}_{-0.0010}$	0.9520 $^{+0.0053}_{-0.0053}$	0.9671 $^{+0.0010}_{-0.0010}$
50–60	0.9285 $^{+0.0089}_{-0.0047}$	0.9498 $^{+0.0010}_{-0.0010}$	0.9476 $^{+0.0010}_{-0.0010}$	0.9576 $^{+0.0039}_{-0.0056}$	0.9679 $^{+0.0010}_{-0.0010}$
60–70	0.9231 $^{+0.0079}_{-0.0080}$	0.9487 $^{+0.0078}_{-0.0108}$	0.9543 $^{+0.0037}_{-0.0010}$	0.9625 $^{+0.0055}_{-0.0078}$	0.9774 $^{+0.0024}_{-0.0030}$
70–90	0.9611 $^{+0.0185}_{-0.0185}$	0.9611 $^{+0.0019}_{-0.0019}$	0.9720 $^{+0.0029}_{-0.0040}$	0.9641 $^{+0.0072}_{-0.0071}$	0.9780 $^{+0.0014}_{-0.0040}$
90– ∞	0.9399 $^{+0.0206}_{-0.0240}$	0.9711 $^{+0.0023}_{-0.0029}$	0.9794 $^{+0.0010}_{-0.0018}$	0.9793 $^{+0.0053}_{-0.0053}$	0.9870 $^{+0.0012}_{-0.0019}$

Table 12: 2023 – postBPix – standard (SS) – unseeded (sublead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–21 †	0.6451 $^{+0.0294}_{-0.0653}$	0.7797 $^{+0.0078}_{-0.0093}$	0.7955 $^{+0.0016}_{-0.0047}$	0.6801 $^{+0.0192}_{-0.0217}$	0.8336 $^{+0.0072}_{-0.0079}$
21–25	0.8564 $^{+0.0350}_{-0.0144}$	0.9069 $^{+0.0017}_{-0.0017}$	0.9005 $^{+0.0088}_{-0.0011}$	0.8754 $^{+0.0177}_{-0.0346}$	0.8825 $^{+0.0072}_{-0.0072}$
25–28	0.8905 $^{+0.0132}_{-0.0362}$	0.9458 $^{+0.0060}_{-0.0043}$	0.9470 $^{+0.0012}_{-0.0012}$	0.8790 $^{+0.0109}_{-0.0159}$	0.9268 $^{+0.0011}_{-0.0011}$
28–31	0.9282 $^{+0.0037}_{-0.0206}$	0.9602 $^{+0.0031}_{-0.0022}$	0.9640 $^{+0.0013}_{-0.0014}$	0.9010 $^{+0.0186}_{-0.0211}$	0.9506 $^{+0.0037}_{-0.0031}$
31–35	0.9457 $^{+0.0057}_{-0.0061}$	0.9696 $^{+0.0017}_{-0.0018}$	0.9677 $^{+0.0010}_{-0.0010}$	0.9405 $^{+0.0102}_{-0.0037}$	0.9659 $^{+0.0010}_{-0.0010}$
35–40	0.9396 $^{+0.0010}_{-0.0010}$	0.9644 $^{+0.0010}_{-0.0010}$	0.9686 $^{+0.0010}_{-0.0010}$	0.9451 $^{+0.0015}_{-0.0015}$	0.9723 $^{+0.0010}_{-0.0010}$
40–45	0.9533 $^{+0.0010}_{-0.0010}$	0.9721 $^{+0.0010}_{-0.0010}$	0.9755 $^{+0.0010}_{-0.0010}$	0.9560 $^{+0.0016}_{-0.0014}$	0.9809 $^{+0.0010}_{-0.0010}$
45–50	0.9580 $^{+0.0010}_{-0.0010}$	0.9752 $^{+0.0010}_{-0.0010}$	0.9784 $^{+0.0010}_{-0.0010}$	0.9628 $^{+0.0025}_{-0.0035}$	0.9840 $^{+0.0010}_{-0.0010}$
50–60	0.9585 $^{+0.0022}_{-0.0022}$	0.9733 $^{+0.0010}_{-0.0011}$	0.9772 $^{+0.0010}_{-0.0010}$	0.9618 $^{+0.0046}_{-0.0055}$	0.9857 $^{+0.0010}_{-0.0013}$
60–70	0.9548 $^{+0.0094}_{-0.0092}$	0.9656 $^{+0.0012}_{-0.0012}$	0.9787 $^{+0.0031}_{-0.0010}$	0.9700 $^{+0.0065}_{-0.0069}$	0.9925 $^{+0.0014}_{-0.0028}$
70–90	0.9645 $^{+0.0088}_{-0.0088}$	0.9741 $^{+0.0016}_{-0.0015}$	0.9841 $^{+0.0023}_{-0.0022}$	0.9691 $^{+0.0041}_{-0.0076}$	0.9891 $^{+0.0010}_{-0.0010}$
90– ∞	0.9405 $^{+0.0134}_{-0.0300}$	0.9751 $^{+0.0020}_{-0.0021}$	0.9830 $^{+0.0010}_{-0.0010}$	0.9746 $^{+0.0045}_{-0.0051}$	0.9928 $^{+0.0010}_{-0.0010}$

Table 13: 2023 – preBPix – fullPath – seeded (lead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwDSCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–35 [†]	0.8618 ^{+0.0070} _{–0.0060}	0.8946 ^{+0.0034} _{–0.0034}	0.8940 ^{+0.0012} _{–0.0012}	0.7977 ^{+0.0192} _{–0.0202}	0.8559 ^{+0.0012} _{–0.0012}
35–37	0.9492 ^{+0.0053} _{–0.0053}	0.9592 ^{+0.0010} _{–0.0011}	0.9365 ^{+0.0010} _{–0.0010}	0.9623 ^{+0.0070} _{–0.0050}	0.9690 ^{+0.0018} _{–0.0017}
37–40	0.9507 ^{+0.0029} _{–0.0029}	0.9650 ^{+0.0010} _{–0.0010}	0.9417 ^{+0.0010} _{–0.0010}	0.9698 ^{+0.0055} _{–0.0053}	0.9734 ^{+0.0010} _{–0.0010}
40–45	0.9605 ^{+0.0021} _{–0.0021}	0.9696 ^{+0.0010} _{–0.0010}	0.9507 ^{+0.0010} _{–0.0010}	0.9776 ^{+0.0025} _{–0.0019}	0.9766 ^{+0.0010} _{–0.0010}
45–50	0.9630 ^{+0.0029} _{–0.0029}	0.9704 ^{+0.0010} _{–0.0010}	0.9571 ^{+0.0010} _{–0.0010}	0.9796 ^{+0.0021} _{–0.0023}	0.9795 ^{+0.0010} _{–0.0013}
50–60	0.9644 ^{+0.0049} _{–0.0046}	0.9731 ^{+0.0057} _{–0.0057}	0.9624 ^{+0.0014} _{–0.0014}	0.9833 ^{+0.0040} _{–0.0040}	0.9793 ^{+0.0030} _{–0.0022}
60–70	0.9815 ^{+0.0091} _{–0.0120}	0.9754 ^{+0.0031} _{–0.0046}	0.9717 ^{+0.0010} _{–0.0010}	0.9947 ^{+0.0025} _{–0.0026}	0.9871 ^{+0.0033} _{–0.0062}
70–90	0.9944 ^{+0.0069} _{–0.0070}	0.9846 ^{+0.0033} _{–0.0034}	0.9821 ^{+0.0011} _{–0.0011}	0.9910 ^{+0.0068} _{–0.0068}	0.9905 ^{+0.0044} _{–0.0132}
90– ∞	0.9274 ^{+0.0321} _{–0.0321}	0.9902 ^{+0.0043} _{–0.0039}	0.9940 ^{+0.0010} _{–0.0010}	0.9984 ^{+0.0010} _{–0.0010}	0.9934 ^{+0.0020} _{–0.0020}

 Table 14: 2023 – preBPix – fullPath – unseeded (sublead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwDSCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–21 [†]	0.9874 ^{+0.0035} _{–0.0036}	0.9914 ^{+0.0021} _{–0.0021}	0.9949 ^{+0.0010} _{–0.0010}	0.9819 ^{+0.0070} _{–0.0112}	0.9702 ^{+0.0107} _{–0.0028}
21–25	0.9966 ^{+0.0011} _{–0.0011}	0.9981 ^{+0.0010} _{–0.0010}	0.9992 ^{+0.0010} _{–0.0010}	0.9940 ^{+0.0036} _{–0.0062}	0.9949 ^{+0.0040} _{–0.0014}
25–28	0.9997 ^{+0.0010} _{–0.0010}	0.9988 ^{+0.0010} _{–0.0010}	0.9987 ^{+0.0010} _{–0.0010}	0.9990 ^{+0.0017} _{–0.0026}	0.9983 ^{+0.0010} _{–0.0010}
28–31	0.9998 ^{+0.0010} _{–0.0010}	0.9994 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9996 ^{+0.0010} _{–0.0010}	0.9991 ^{+0.0010} _{–0.0010}
31–35	0.9997 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9993 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
35–40	0.9998 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	1.0000 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}
40–45	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
45–50	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
50–60	0.9997 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	1.0000 ^{+0.0010} _{–0.0010}
60–70	0.9965 ^{+0.0013} _{–0.0013}	0.9999 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9989 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
70–90	0.9944 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9977 ^{+0.0021} _{–0.0021}	0.9997 ^{+0.0010} _{–0.0010}
90– ∞	0.9723 ^{+0.0085} _{–0.0085}	0.9992 ^{+0.0014} _{–0.0014}	0.9996 ^{+0.0010} _{–0.0010}	0.9984 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}

 Table 15: 2023 – postBPix – fullPath – seeded (lead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwDSCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–35 [†]	0.7994 ^{+0.0072} _{–0.0142}	0.8130 ^{+0.0028} _{–0.0029}	0.8664 ^{+0.0010} _{–0.0010}	0.8418 ^{+0.0093} _{–0.0058}	0.8927 ^{+0.0022} _{–0.0021}
35–37	0.9455 ^{+0.0021} _{–0.0021}	0.9546 ^{+0.0014} _{–0.0010}	0.9340 ^{+0.0010} _{–0.0010}	0.9594 ^{+0.0088} _{–0.0037}	0.9706 ^{+0.0010} _{–0.0016}
37–40	0.9494 ^{+0.0016} _{–0.0016}	0.9619 ^{+0.0010} _{–0.0010}	0.9418 ^{+0.0010} _{–0.0010}	0.9718 ^{+0.0027} _{–0.0020}	0.9749 ^{+0.0010} _{–0.0010}
40–45	0.9544 ^{+0.0010} _{–0.0010}	0.9670 ^{+0.0010} _{–0.0010}	0.9502 ^{+0.0010} _{–0.0010}	0.9786 ^{+0.0010} _{–0.0010}	0.9773 ^{+0.0010} _{–0.0010}
45–50	0.9598 ^{+0.0024} _{–0.0024}	0.9690 ^{+0.0010} _{–0.0010}	0.9565 ^{+0.0010} _{–0.0010}	0.9834 ^{+0.0010} _{–0.0012}	0.9788 ^{+0.0010} _{–0.0010}
50–60	0.9625 ^{+0.0037} _{–0.0032}	0.9715 ^{+0.0010} _{–0.0026}	0.9640 ^{+0.0010} _{–0.0010}	0.9851 ^{+0.0025} _{–0.0036}	0.9771 ^{+0.0010} _{–0.0010}
60–70	0.9654 ^{+0.0105} _{–0.0080}	0.9768 ^{+0.0024} _{–0.0016}	0.9731 ^{+0.0010} _{–0.0010}	0.9868 ^{+0.0071} _{–0.0072}	0.9827 ^{+0.0020} _{–0.0022}
70–90	0.9799 ^{+0.0094} _{–0.0080}	0.9841 ^{+0.0019} _{–0.0020}	0.9839 ^{+0.0010} _{–0.0010}	0.9907 ^{+0.0036} _{–0.0036}	0.9879 ^{+0.0013} _{–0.0012}
90– ∞	0.9928 ^{+0.0020} _{–0.0048}	0.9918 ^{+0.0014} _{–0.0015}	0.9937 ^{+0.0010} _{–0.0010}	0.9990 ^{+0.0010} _{–0.0010}	0.9928 ^{+0.0014} _{–0.0010}

Table 16: 2023 – postBPix – fullPath – unseeded (sublead) leg. Nominal $\varepsilon_{-\sigma-}^{+\sigma+}$ (bwD-SCB $\otimes$ CMSShape).

E_T [GeV]	EB R_9 0.50–0.55	EB R_9 0.55–0.85	EB $R_9 > 0.85$	EE R_9 0.90–0.93	EE $R_9 > 0.93$
0–21 [†]	0.9858 ^{+0.0052} _{–0.0052}	0.9891 ^{+0.0040} _{–0.0026}	0.9898 ^{+0.0018} _{–0.0018}	0.9823 ^{+0.0105} _{–0.0151}	0.9877 ^{+0.0018} _{–0.0018}
21–25	0.9960 ^{+0.0024} _{–0.0018}	0.9966 ^{+0.0010} _{–0.0010}	0.9965 ^{+0.0010} _{–0.0010}	0.9917 ^{+0.0026} _{–0.0027}	0.9948 ^{+0.0010} _{–0.0010}
25–28	0.9988 ^{+0.0010} _{–0.0010}	0.9986 ^{+0.0010} _{–0.0011}	0.9986 ^{+0.0010} _{–0.0010}	0.9972 ^{+0.0028} _{–0.0028}	0.9977 ^{+0.0010} _{–0.0010}
28–31	0.9987 ^{+0.0011} _{–0.0010}	0.9994 ^{+0.0010} _{–0.0010}	0.9989 ^{+0.0010} _{–0.0010}	0.9986 ^{+0.0013} _{–0.0010}	0.9992 ^{+0.0010} _{–0.0010}
31–35	0.9994 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9994 ^{+0.0010} _{–0.0010}	0.9993 ^{+0.0010} _{–0.0010}
35–40	0.9996 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
40–45	0.9996 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
45–50	0.9997 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
50–60	0.9998 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}
60–70	0.9994 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9994 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
70–90	0.9988 ^{+0.0022} _{–0.0022}	1.0000 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9998 ^{+0.0010} _{–0.0010}
90– ∞	0.9961 ^{+0.0010} _{–0.0010}	0.9991 ^{+0.0010} _{–0.0010}	0.9997 ^{+0.0010} _{–0.0010}	0.9995 ^{+0.0010} _{–0.0010}	0.9999 ^{+0.0010} _{–0.0010}

9 Provenance & status

- **Fitter:** `egm.tnp.analysis` under `CMSSW_10.6.8` on `lxplus`; driver `run_HLT_2024_all.sh`.
- **2024:** all four variants complete (40 category $\times$ leg combinations, both fit models). The three unseeded categories that first returned non-converged dummy values, and the entire `fullPath` variant, were re-fit on `lxplus` (2026-07). No dummy (0.5 ± 0.001) values remain.
- **2023:** standard and `fullPath`, both eras, all categories; includes HiggsDNA TriggerSF correctionlib JSONs.
- **Not included:** HiggsDNA TriggerSF JSONs for 2024 (not yet built).
- **Reproducibility:** CSV + plots from `2026-07-14_build_eff_csv_and_plots.ipynb`; repo staging from `2026-07-14_build_repo_artifacts.py`; this document from `build_production.pdf.py`.